

Bacterial leaf blight (BLB) is a serious bacterial disease of rice caused by *Xanthomonas oryzae* pv. *oryzae* (**Xoo**). It infects rice plants through wounds or natural openings such as hydathodes, then spreads through the plant's water-conducting tissues (xylem). The disease causes wilting of seedlings, yellowing and drying of leaves, and can lead to severe yield losses, especially when infection occurs early in crop growth.

Bacterial leaf blight (BLB) can be recognized by **yellowing, drying, and wilting of leaves**. In young seedlings, leaves may first turn **grayish-green, roll up, then become yellow to straw-colored and dry out** (known as *kresek* symptoms). On older plants, the disease usually appears as **water-soaked to yellow-orange stripes along leaf margins or tips**, which gradually extend downward. Early in the morning, infected leaves may show **milky bacterial ooze droplets** that later dry into **small yellowish beads**. As the disease progresses, lesions may turn **yellow to grayish-white**, and severely infected leaves can dry completely.

Yellowing and drying starting from the leaf tips is a common symptom of **Bacterial Leaf Blight (BLB)** caused by *Xanthomonas oryzae* pv. *oryzae*. Infected leaves may first show water-soaked or yellow-orange streaks that begin near the leaf tip or leaf margin and then spread downward. As the disease progresses, leaves turn yellow, dry out, and may appear blighted.

Leaf Scald can also begin at the leaf tip, with lesions extending downward from the tip. Severely affected leaves eventually dry and become straw-colored.

The earliest signs of bacterial leaf blight (BLB) depend on the crop stage. In seedlings, the first symptoms are **wilting**, leaves turning **grayish-green**, rolling up, then becoming **yellow to straw-colored** before drying. In older plants, the disease usually begins as **water-soaked to yellow-orange streaks** near the leaf tip or leaf margins, especially on injured leaves. Young lesions may also produce a **milky bacterial ooze** in the early morning.

BLB is described as causing vascular wilt in early growth stages and leaf blight later in crop development. The IRRI factsheet states that seedlings show wilting and grayish-green rolled leaves, while older plants develop water-soaked yellow-orange streaks and bacterial ooze.

These symptoms help distinguish BLB from many fungal leaf diseases, which usually begin as discrete spots rather than water-soaked streaks.

Farmer-friendly summary:

Look for **yellowing and wilting leaves**, especially starting from the leaf tip or edges. Early BLB lesions often appear as **water-soaked yellow streaks**, and in the morning you may see

sticky, milky bacterial droplets (ooze) on young lesions. These are among the earliest warning signs of bacterial leaf blight.

Bacterial Leaf Blight (BLB) usually starts as water-soaked to yellow-orange stripes along leaf tips or leaf margins. The lesions extend downward along the leaf blade, causing yellowing, drying, and wilting of leaves. Young lesions may produce a milky bacterial ooze that later dries into yellowish beads. Seedlings can show a severe wilting phase called **kresek**.

If you cut a fresh BLB lesion and place it in clean water, bacterial streaming/ooze may be visible, which does not occur with leaf blast. This diagnostic method is described in the IRRI BLB factsheet

Drying of leaf margins is a characteristic symptom of **Bacterial Leaf Blight (BLB)** caused by *Xanthomonas oryzae* pv. *oryzae*. According to the IRRI factsheet, lesions begin as water-soaked to yellow-orange stripes near leaf tips or margins and then enlarge downward along the leaf. As the disease progresses, affected leaf tissue turns yellow, dries, and may appear scorched from the margins inward.

Bacterial Leaf Blight (BLB) can infect young rice plants and seedlings. The disease may appear as **kresek**, a severe seedling-stage symptom where leaves wilt, turn yellow to straw-colored, dry up, and the entire seedling may die. Early infection is particularly damaging and can cause major yield losses or even crop failure.

Kresek is the seedling phase of bacterial leaf blight (BLB). Infected seedlings show **wilting**, leaves become **grayish-green and rolled**, then turn **yellow to straw-colored**, dry up, and may die. Kresek symptoms can resemble stem borer damage, but in kresek, a cut infected seedling may release **yellowish bacterial ooze** and the plant is usually not easily pulled from the soil.

Sudden wilting of rice leaves can be a symptom of **Bacterial Leaf Blight (BLB)** caused by *Xanthomonas oryzae* pv. *oryzae*. The disease causes wilting of seedlings (known as **kresek**) and yellowing and drying of leaves. The bacterium enters through wounds or natural openings (hydathodes), multiplies inside the water-conducting xylem vessels, and blocks water movement, leading to wilting.

Bacterial leaf blight (BLB) of rice is caused by the bacterium **Xanthomonas oryzae pv. oryzae (Xoo)**. The bacterium enters rice plants through natural openings such as hydathodes or through wounds, then colonizes the xylem (water-conducting tissues), causing wilting, yellowing, and blighting of leaves.

The IRRI fact sheet also states that bacterial blight is caused by **Xanthomonas oryzae pv. oryzae** and causes wilting of seedlings and yellowing and drying of leaves.

The Gazipur Agricultural University rice disease guide lists **Bacterial blight – *Xanthomonas oryzae* pv. *oryzae*** under bacterial diseases of rice

Bacterial Leaf Blight (BLB) of rice is caused by the bacterium ***Xanthomonas oryzae* pv. *oryzae* (Xoo)**. This bacterium infects rice plants through wounds or natural openings (hydathodes) and spreads through the xylem (water-conducting tissue), causing leaf blight and yield loss.

Bacterial Leaf Blight (BLB) spreads through several routes. The pathogen (*Xanthomonas oryzae* pv. *oryzae*) can survive in infected seeds, stems, stubbles, roots, and other plant parts after harvest. It is spread by rain splash, irrigation water, birds, insects, and human activities. The bacteria enter rice plants through natural openings (hydathodes) or wounds caused by insects or mechanical damage, then multiply and spread through the xylem (water-conducting tissues).

BLB is also favored by strong winds and continuous heavy rains, which help bacteria move through water droplets from infected plants to healthy plants. Weeds, infected stubble, ratoons, and volunteer rice plants can act as sources of infection.

Irrigation water can spread Bacterial Leaf Blight (BLB). The BLB review paper identifies irrigation water as one of the important sources of the pathogen, and recommends managing these sources to reduce disease incidence. The pathogen can also spread through rain splash and contaminated water movement in fields.

If water flows from a field infected with BLB into a healthy field, it can help move the bacteria and increase the chance of disease spread. Good drainage, clean irrigation sources, and avoiding movement of contaminated water between fields can help reduce risk.

The uploaded documents state that bacterial leaf blight (BLB) can spread through rain. The pathogen can be dispersed by **rain splash**, and the IRRI factsheet specifically notes that **strong winds together with continuous heavy rain** help spread the bacteria through water droplets and lesions on infected plants.

The uploaded BLB review paper states that *rain splash, birds, insects, and human activities can spread* the bacterial leaf blight (BLB) pathogen, ***Xanthomonas oryzae* pv. *oryzae* (Xoo)**. The pathogen can also survive in infected crop residues and alternate hosts, which helps its spread between fields.

The uploaded BLB review paper states that *Xanthomonas oryzae* pv. *oryzae* (the bacterium causing bacterial leaf blight) can infect the **seed, stem, and other plant parts of rice even after harvest**, indicating that seed-associated infection can occur.

Seed-borne survival of BLB bacteria has been reported in plant pathology literature, but infected crop residues, volunteer plants, and contaminated water are generally considered more important sources for field epidemics than seed transmission alone.

Bacterial Leaf Blight (BLB) outbreaks are favored by **warm temperatures, high humidity, monsoon conditions, heavy rainfall, and strong winds**. The disease is most common when temperatures are around **25–34°C** and relative humidity is **above 70%**. Continuous rains and strong winds help spread the bacteria through water droplets and wounds on rice plants. The BLB review paper also notes that disease occurrence is highest during the **monsoon season (June–September in India)**.

Bacterial Leaf Blight (BLB) often increases after storms because strong winds and heavy rains create wounds on rice leaves and help spread the bacteria. The pathogen (*Xanthomonas oryzae* pv. *oryzae*) can enter the plant through wounds and natural openings (hydathodes). Rain splash, wind-driven water droplets, insects, birds, and other activities can spread the bacteria from infected plants to healthy plants. Storm conditions therefore increase both bacterial dispersal and infection opportunities.

Excessive nitrogen fertilization can increase the incidence and severity of bacterial leaf blight (BLB). The uploaded BLB review specifically recommends **avoiding excessive nitrogen fertilizer** as part of BLB management. The IRRI factsheet also states that bacterial blight can become severe in susceptible rice varieties under **high nitrogen fertilization**.

an integrated approach is recommended. Key management practices include: **(1) growing resistant rice varieties**, which the paper describes as the most effective strategy; **(2) cultural practices** such as good drainage, complete drying of fallow fields before sowing, avoiding overhead irrigation, timely weed control, avoiding excessive nitrogen fertilizer, reducing dense crop canopy, aerating soil, and crop rotation; and **(3) biological control agents (BCAs)** such as *Pseudomonas*, *Bacillus*, and *Trichoderma*, which suppress the pathogen through antimicrobial compounds, competition, and induction of plant defenses. The paper also notes that biological control can complement chemical treatments and contribute to sustainable disease management.

The IRRI factsheet additionally recommends: **plant resistant varieties**, use **balanced plant nutrients (especially avoid excessive nitrogen)**, ensure **good field drainage**, keep fields clean by removing weeds, stubble, ratoons, and volunteer rice plants, and allow fallow fields to dry to suppress disease sources.

chemical products such as streptocycline, streptomycin sulfate, copper oxychloride, kasugamycin, and copper hydroxide have been used against BLB. However, the paper also highlights concerns regarding pathogen resistance, environmental impacts, and effects on beneficial organisms, which is why biological control and resistant varieties are receiving increased attention.

The uploaded BLB review paper reports that several bactericidal/antibacterial chemicals have been used against bacterial leaf blight (BLB). **Streptomycin sulfate** showed high effectiveness (92.23%) against BLB, while **copper oxychloride** showed 76.48% effectiveness compared with untreated controls. The paper also mentions **streptocycline** (containing

streptomycin sulfate and tetracycline hydrochloride), **kasugamycin**, **copper hydroxide**, and combinations such as **streptocycline + copper oxychloride** for BLB management.

The documents indicate that antibiotics have been used against bacterial leaf blight (BLB), but they should not be considered the primary or most sustainable management strategy.

The review paper reports the use of antibiotics such as **streptomycin sulfate**, **tetracycline**, **streptocycline**, **kasugamycin**, and **agrimycin** for BLB suppression. However, it also highlights concerns about the development of bacterial resistance, environmental impacts, and risks to beneficial microorganisms. The paper emphasizes resistant varieties, cultural practices, and biological control as more sustainable approaches.

The IRRI factsheet does **not recommend antibiotics** as a routine management practice. Instead, it recommends **planting resistant varieties, balanced fertilizer use (especially avoiding excess nitrogen), good drainage, field sanitation, removal of infected residues, and drying fallow fields.**

The reviewed research paper reports that several copper-based products have shown effectiveness against bacterial leaf blight (BLB). Copper oxychloride showed **76.48% effectiveness** in BLB management compared with untreated controls. Copper hydroxide (77% WP) showed up to **78.57% control efficiency** after inoculation. Copper nanoparticles (0.2%) also inhibited the BLB pathogen (*Xanthomonas oryzae* pv. *oryzae*) in laboratory tests. In addition, combining copper oxychloride with streptocycline reduced disease incidence more than untreated controls.

chemicals such as streptocycline, copper oxychloride, copper hydroxide, and related compounds have been used against BLB, but it does not specify an exact crop stage for routine spraying.

For bacterial leaf blight, sprays are generally most effective when applied **immediately after the first appearance of disease symptoms** or when weather conditions strongly favor disease development (frequent rains, strong winds, high humidity). Preventive management through resistant varieties, balanced nitrogen use, field sanitation, and proper drainage is usually emphasized more than routine calendar-based spraying.

In practice, application frequency should follow the label instructions of the specific biocontrol product, bactericide, or fungicide being used, and local agricultural extension recommendations. Repeated applications may be needed during periods favorable for disease development, but no single frequency can be recommended from the uploaded documents alone.

For severe **bacterial leaf blight (BLB)** infection, the documents recommend an integrated approach: **plant resistant varieties**, maintain **balanced fertilizer use (especially avoid excessive nitrogen)**, ensure **good field drainage**, remove infected crop residues, weeds, ratoons, and volunteer rice plants, and allow fields to dry between crops when possible. The

review paper also notes that biological control agents (such as *Pseudomonas*, *Bacillus*, and *Trichoderma*) and certain chemical treatments have been used as part of disease management programs.

Partially, depending on when and how severely the plant is infected. In bacterial leaf blight (BLB), early infection can cause very severe damage and may result in major yield loss or even crop failure during the tillering stage. Plants infected later in the season may survive but can still suffer reduced grain quality and yield. Once leaf tissue has been killed by the disease, the damaged tissue does not recover, although unaffected parts of the plant may continue to grow if conditions are favorable.

Rice plants cannot regenerate dead leaf tissue damaged by BLB. However, if infection is mild and occurs late, healthy leaves and tillers may continue functioning, allowing the plant to produce some yield. Recovery is therefore limited and depends on disease severity, crop stage, variety, and management.

focusing on measures that reduce sources of infection and limit spread. Quickly remove or manage infected plant residues, weeds, ratoons, and volunteer rice plants that can harbor the bacteria. Maintain good field drainage, avoid overhead irrigation that keeps leaves wet, and avoid excessive nitrogen fertilizer because it can increase disease severity. Planting resistant varieties is considered the most effective and reliable management strategy.

If disease symptoms are already widespread in the field, complete eradication is usually not possible during the current crop cycle. The goal becomes slowing further spread and protecting healthy plants through sanitation, water management, balanced nutrition, and resistant varieties in future plantings.

To reduce the risk of Bacterial Leaf Blight (BLB) in future rice crops, use an integrated approach: **(1)** plant BLB-resistant rice varieties whenever available, **(2)** avoid excessive nitrogen fertilizer application, **(3)** maintain good field drainage, **(4)** remove infected crop residues, volunteer rice plants, ratoons, and weed hosts, **(5)** allow fallow fields to dry before planting, and **(6)** avoid practices that keep leaves excessively wet. The reviewed research also identifies resistant varieties as one of the most effective management strategies and recommends cultural practices that reduce sources of the pathogen.

For **bacterial leaf blight (BLB)**, the reviewed research paper does **not provide specific variety names**, but states that rice varieties carrying resistance genes such as **Xa4, xa5, xa7, xa13, xa21, xa23, and xa33** have been successfully developed and that combining (pyramiding) these genes provides broad-spectrum resistance against BLB.

The specific variety to plant depends on your country and region because BLB-resistant varieties differ by national breeding programs. Farmers should select varieties officially recommended by their local agricultural extension service or rice research institute that contain BLB resistance genes (especially combinations involving Xa21, xa13, xa5, xa7, etc.).

Field sanitation is recommended as an important measure to reduce Bacterial Leaf Blight (BLB). This includes removing infected plant material, rice stubble, straw, ratoons, volunteer seedlings, and weeds that can serve as sources of the BLB bacterium. Good field hygiene helps reduce the amount of pathogen present in the field and lowers disease spread.

Bacterial Leaf Blight (BLB) can cause **up to 74% yield loss** under severe disease conditions. The disease is particularly destructive when infection occurs early in crop growth. The review paper also reports country-specific losses ranging from **20–80% in India, 20–40% in Bangladesh, 30–50% in the Philippines, up to 50% in Malaysia, and 50–80% in Mali.**

Bacterial Leaf Blight (BLB) can affect grain filling, especially when infection occurs early or before the grain-filling stage. The disease damages the plant's water-conducting tissues (xylem), reducing water and nutrient movement, which can lead to poor grain development and yield loss. The IRRI factsheet also notes that when infection occurs at the booting stage, grain quality declines and the proportion of broken kernels increases.

Bacterial Leaf Blight (BLB) can reduce grain quality. According to the IRRI Rice Knowledge Bank factsheet, when rice plants are infected at the **booting stage**, BLB may not significantly reduce yield, but it can result in **poor-quality grains and a high proportion of broken kernels.**

If Bacterial Leaf Blight (BLB) is not controlled, it can cause severe yield losses, poor grain quality, and in severe early infections even complete crop failure. The disease causes wilting of seedlings, yellowing and drying of leaves, blockage of water-conducting tissues (xylem), and reduced plant vigor. Early infection can destroy up to 80% of yield, while infections during the tillering stage may lead to total crop failure. BLB is considered one of the most serious rice diseases because yield loss increases when infection occurs early in crop growth.

BLB causes wilting of seedlings and yellowing and drying of leaves. It identifies BLB as one of the most serious rice diseases and notes that earlier disease occurrence results in greater yield loss.

Bacterial Leaf Blight (BLB) can spread throughout a field when conditions are favorable.

The pathogen can be spread by rain splash, irrigation water, insects, birds, and human activities. Once established, the bacteria enter rice plants through wounds or natural openings and multiply within the water-conducting tissues (xylem), leading to systemic infection. The disease can also survive on infected plant material, stubble, and alternative hosts, allowing further spread within and between fields.